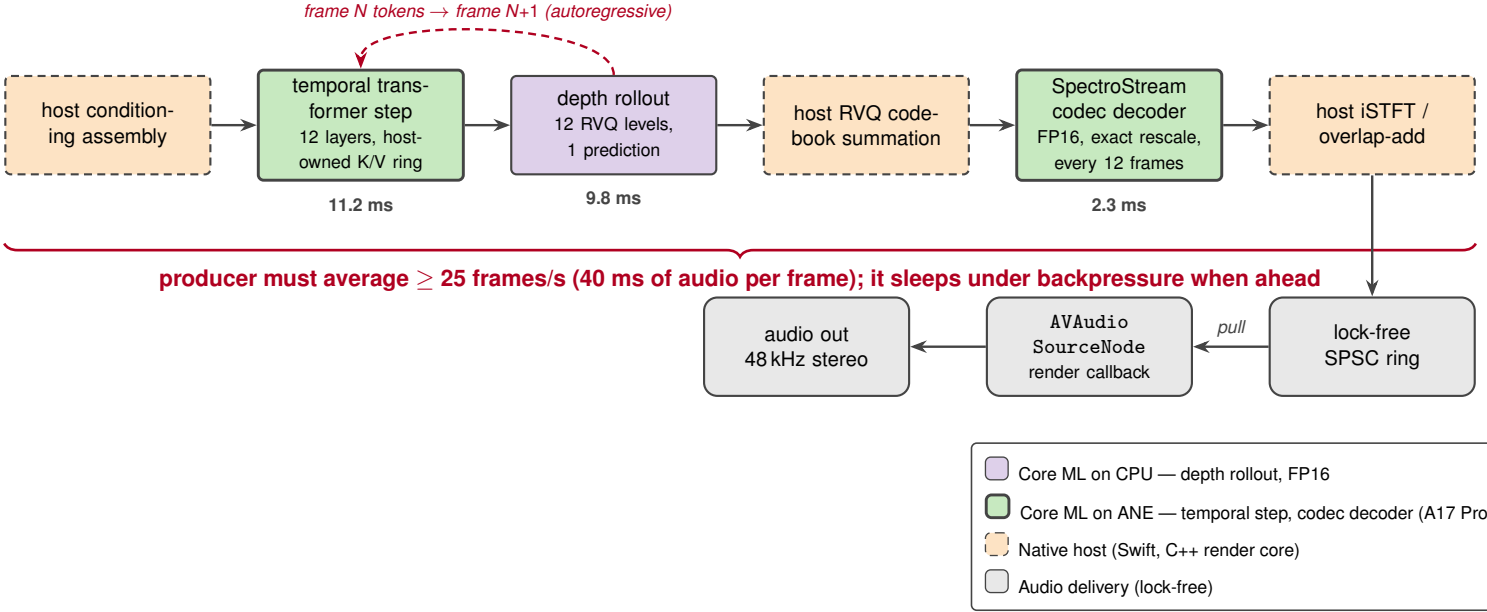
frame N tokens → frame N+1 (autoregressive)
host condition-ing assembly
temporal trans-former step
12 layers, host-owned K/V ring
11.2 ms
depth rollout
12 RVQ levels, 1 prediction
9.8 ms
host RVQ code-book summation
SpectroStream codec decoder
FP16, exact rescale, every 12 frames
2.3 ms
host iSTFT / overlap-add
producer must average ≥ 25 frames/s (40 ms of audio per frame); it sleeps under backpressure when ahead
lock-free SPSC ring
pull
AVAudio SourceNode
render callback
audio out
48 kHz stereo
Core ML on CPU — depth rollout, FP16
Core ML on ANE — temporal step, codec decoder (A17 Pro)
Native host (Swift, C++ render core)
Audio delivery (lock-free)